Vinod A. Thale

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Vinod A. Thale | Vinod A. Thale
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Profile

- **Fluid mechanics researcher, graduated from Xi'an Jiaotong University with a research focus on multiphase computational fluid dynamics (CFD), particularly interfacial flows and drop dynamics. Demonstrated expertise in the Volume-of-Fluid (VOF) method using the Basilisk open-source solver, with additional proficiency in OpenFOAM, ANSYS Fluent, and COMSOL Multiphysics. Research work includes direct numerical simulations of compound drop impact, counter-jet formation, and transient force modeling. Experienced in simulation validation and collaborative scientific investigations across multidisciplinary domains.**

Education

2018-07 – 2024-07
- **Ph.D. in Mechanics, Xi'an Jiaotong University, Xi'an, China. School of Aerospace Engineering (CGPA: *3.74/4*).**
  Thesis Title: *Numerical Modelling of an Air-in-Liquid Compound Drop*.
  Advisor: **Prof. Marie-Jean THORAVAL**
  Description: *Focused on the dynamics of hollow drops, with applications in plasma spraying of thermal barrier coatings and sea spray aerosols. Investigated force balance and counter-jet phenomena using theoretical models and the open-source CFD tool Basilisk. Performed direct numerical simulations on HPC clusters.*

2017-09 – 2018-06
- **Chinese Language Course, Xi'an Jiaotong University, Xi'an, China.** *HSK Level 3 certified.*

2013-07 – 2015-07
- **M.Tech. in Computational Fluid Dynamics, UPES Dehradun (CGPA: *3.02/4*).**
  Thesis Title: *Incompressible Flow Simulation in a Channel with a Square Cylinder Using FORTRAN.*
  Advisor: **Dr. K.P. Singh, ADA Bangalore**
  Description: *Studied the impact of channel aspect ratio on square cylinder wakes using exit velocity profiles at various inlet distances. Developed an in-house FORTRAN flow solver validated against cavity benchmarks, with results compared to experimental and numerical data.*

2006-07 – 2010-07
- **Bachelor in Mechanical Engineering, North Maharashtra University, Jalgaon (Percentage: *68.20%*, CGPA: *2.73/4.00*).**
  Thesis Title: *Flow Analysis in Refrigerator Tubing.*

2006 – 2007
- **Higher Secondary Certificate (HSC), Maharashtra State Board. (Percentage: *65.50%*, CGPA: *2.62/4.00*).**

2004 – 2005
- **Secondary School Certificate (SSC), Maharashtra State Board. (Percentage: *77.20%*, CGPA: *3.09/4.00*).**

Employment History

2025-04 – 2025-07 **Research Associate.** **Indian Institute of Technology Hyderabad, India**
Department: *Department of Mechanical and Aerospace Engineering, and Department of Chemical Engineering*
Project title: *Non-spherical aerodynamic breakup of droplets using DNS.*
Advisors: ***Prof. Sachidananda Behera*** *(Mechanical and Aerospace Engineering),* ***Prof. Kirti Chandra Sahu*** *(Chemical Engineering)*
Responsibilities: *Conducting high-fidelity direct numerical simulations (DNS) of aerodynamic droplet breakup and analyzing breakup regimes under varying Weber and Reynolds numbers. Utilizing Basilisk open-source CFD software for grid-adaptive multiphase flow simulations.*

2024-07 – 2025-03 **Product Development Engineer (CFD).** **Guangdong Lingyi iTECH Manufacturing Co. Ltd, Shenzhen, China**
Responsibilities: *CFD simulation and optimization of a vapor chamber using ANSYS Fluent and the Basilisk open-source CFD software. Simulation and validation of the thermal design for a vapor chamber in a phone product. Writing UDFs for vapor chamber modeling.*

2016-10 – 2017-06 **Junior Research Fellow.** **Indian Institute of Technology Bombay, India**
Advisor: ***Prof. Amitabh Bhattacharya***
Advisor: ***Prof. Tuan TRAN*** **National University of Singapore**
Responsibilities: *Research on nucleate boiling on passive and active flexible microstructured surfaces using an in-house CFD solver. Debugging and verification of the two-phase simulation against theoretical solutions. Conducting systematic numerical simulations and experiments using high-speed imaging.*

2015-08 – 2016-09 **Junior Research Fellow.** **Defence Institute of Advanced Technology**
Advisor: ***Dr. D. Srikanth***
Responsibilities: *CFD two-phase simulation for Non-Newtonian flow, data processing using MATLAB and GNUplot, Tecplot post-processing.*

2010-08 – 2013-06 **Production Engineer.** **Bharat Forge Ltd., Pune, India.** **Bharat Forge**
Responsibilities: *Performed and delivered structural analysis using finite element analysis. Conducted FEA and proposed optimization solutions. Acted as a bridge between the die design and production engineering departments. Contributed to the development and documentation of structural simulation methodologies. Supported designers in establishing certification and validation plans.*

Research Publications

Journal Articles

1. **Thale, Vinod A**, M. Abouelsoud, and M.-J. Thoraval, "Impact forces of drop onto liquid pool: Modeling," *to be submitted*, 2025.
2. **Thale, Vinod A**, M. Abouelsoud, and M.-J. Thoraval, "Scaling of maximum force of a drop," *to be submitted*, 2025.
3. **Thale, Vinod A** and M.-J. Thoraval, "Jet formation during impact of air-in-liquid compound drop impact on pool," *to be submitted*, 2025.
4. **Thale, Vinod A**, M. Abouelsoud, C. . Hossain, and M.-J. Thoraval, "Impact force of an air-in-liquid compound drop," *Physics of Fluids*, vol. 34, no. 1, p. 112 108, Jan. 2024. DOI: 10.1063/5.0183822.
5. M. Abouelsoud, **Thale, Vinod A.**, A. N. Shmroukh, and B. Bai, "Spreading and retraction of the concentric impact of a drop with a sessile drop of the same liquid: Effect of surface wettability," *Physics of Fluids*, vol. 34, no. 11, p. 112 108, Nov. 2022. DOI: 10.1063/5.0117964.

Conference Proceedings

1. **Thale, Vinod A** and M.-J. Thoraval, "Jetting from compound drop," in *: In: APS-DFD (Virtual) (2020)*, 2020.
2. **Thale, Vinod A** and M.-J. Thoraval, "Jet from drop impacting on pool," in *: Basilisk/Gerris Users' meeting, Paris, France (2019) (virtual)*, 2019.

Skills

Spoken Languages	Strong reading, writing, and speaking competencies in English, Chinese, Marathi, and Hindi.
Coding	C, C++, MATLAB, Python, R, GNUPLOT, Octave, LaTeX, ...
CFD software	Ansys-fluent, Comsol, Stat CCM++ , Basilisk, Gerris , Open foam
CAD software	AutoCAD, Solid work
Lab equipment	LabView: Voltage, current module, Conductivity, optical probes, High-speed camera imaging, image processing
Misc.	Academic research, teaching, training, consultation, LaTeX typesetting, and publishing.

Awards and Achievements

2018	**Chinese government scholarship**, Xian Jiotong University, China Xian.
2016	**Junior Research Fellowship**, Defence Institute of Advanced Technology, Government of India, Pune.
2017	**Junior Research Fellowship**, Defence Research and Development Organization DRDO, Government of India, Nagar, Pune.
2016	**Junior Research Fellowship**, Indin Institute Technology, Dhanbad, India.
	Junior Research Fellowship, Indin Institute Technology, Bombay, India.

References

Prof Marie-Jean THORAVAL,
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École Polytechnique, Paris, France ,
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